



QUESTION BANK (AY: 2026-2027) ODD SEMESTER

Department: CSE

Year/ Semester: III / V

Regulation: 2021

Subject Code: CB3491

Subject Name: Cryptography and Cyber Security

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UNIT I INTRODUCTION TO SECURITY

9

Computer Security Concepts – The OSI Security Architecture – Security Attacks – Security Services and Mechanisms – A Model for Network Security – Classical encryption techniques: Substitution techniques, Transposition techniques, Steganography – Foundations of modern cryptography: Perfect security – Information Theory – Product Cryptosystem – Cryptanalysis.

CO1: Understand the fundamentals of networks security, security architecture, threats and vulnerabilities

Two Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	What are Security attack, Security mechanism and Security service? Any action that attempts to compromise the confidentiality, integrity, or availability of information	K1	CO1
2	What is the difference between Monoalphabetic and polyalphabetic cipher? (Dec 2021) [Nov/Dec 2021] Monoalphabetic Cipher • Uses a single substitution alphabet. • Each plaintext letter maps to only one ciphertext letter. • Easier to break using frequency analysis. • Example: Caesar Cipher. Polyalphabetic Cipher • Uses multiple substitution alphabets. • Same plaintext letter can be encrypted differently. • More resistant to frequency analysis. • Example: Vigenère Cipher.	K1	CO1
3	Define confidentiality. Confidentiality is the security service that ensures information is accessible only to authorized users and prevents unauthorized disclosure of data.	K1	CO1
4	Distinguish between attack and threat. [Apr/May 2024] Threat: A potential cause of an unwanted incident that may exploit vulnerabilities. Attack: An actual attempt to exploit a vulnerability and compromise system security.	K1	CO1
5	List 4 general categories of attack. 1. Interruption 2. Interception 3. Modification 4. Fabrication	K1	CO1
6	Differentiate between Interruption and Interception. Interruption • Affects Availability.	K2	CO1

S. No	Question	Knowledge Level	Course Outcome
	- Resource becomes unavailable. - Example: Denial of Service attack. Interception - Affects Confidentiality. - Unauthorized access to information. - Example: Packet sniffing.		
7	Define Authentication, Nonrepudiation, Availability and Access control. Authentication: Verifies the identity of a user or system. Non-repudiation: Prevents a sender or receiver from denying a performed action. Availability: Ensures resources are accessible whenever required. Access Control: Restricts resource access to authorized users only.	K1	CO1
8	Compare active and passive attack Passive Attack - Monitors communication only. - No data modification. - Difficult to detect. - Example: Eavesdropping. Active Attack - Modifies or disrupts data. - Easier to detect. - Example: Masquerade, Replay attack, DoS attack.	K1	CO1
9	Differentiate between Modification and Fabrication. Modification: Unauthorized alteration of existing data affecting integrity. Example: Changing bank account balance. Fabrication: Insertion of fake or counterfeit data into the system affecting authenticity. Example: Sending forged emails.	K2	CO1
10	How cryptographic systems are generally classified? Substitution technique is a classical encryption method in which each plaintext character is replaced by another character, symbol, or number according to a predefined rule while preserving the character positions.	K1	CO1
11	What do you mean by substitutional technique? Substitution technique is a classical encryption method in which each plaintext character is replaced by another character, symbol, or number according to a predefined rule while preserving the character positions.	K1	CO1
12	List some substitution techniques - Caesar Cipher - Playfair Cipher - Hill Cipher - Vigenère Cipher - One-Time Pad	K1	CO1
13	Calculate the ciphertext for the following using one-time pad cipher. Plaintext: R O C K Keyword: B O T S Answer: Using $A=0, B=1, \dots, Z=25$ $R(17)+B(1)=18=S$ $O(14)+O(14)=28 \bmod 26=2=C$ $C(2)+T(19)=21=V$ $K(10)+S(18)=28 \bmod 26=2=C$ Ciphertext = SCVC	K1	CO1
14	Explain the working mechanism of one-time pad. - A truly random key equal in length to the plaintext is generated. - Plaintext and key are combined using modulo-26 addition (or XOR for binary data).	K1	CO1

S. No	Question	Knowledge Level	Course Outcome
	• Ciphertext is transmitted. • Receiver decrypts using the same key.		
15	Compare steganography and cryptography. Cryptography • Hides the meaning of data. • Ciphertext is visible. • Uses encryption algorithms. Steganography • Hides the existence of data. • Secret message is embedded inside media files. • Uses images, audio, or video as cover objects.	K2	CO1
16	Connect the given text “anna university” into cipher text using Rail fence technique. Using 2-rail Rail Fence : Rail 1: A N A U I E S T Rail 2: N A N V R I Y Ciphertext = ANAUIESTNANVRIY	K2	CO1
17	Encrypt the plaintext tobeornottobe using the vigenere cipher for the key value Now. Repeat key: NOWNOWNOWNOWN Apply Vigenère encryption (mod 26). Ciphertext = GCXSKFBBHHCP	K2	CO1
18	How substitution cipher is different from transposition cipher. Give example for each method [Nov/Dec 2021] Substitution Cipher: Characters are replaced while positions remain unchanged. Example: Caesar Cipher. Transposition Cipher: Characters remain the same but their positions are rearranged. Example: Rail Fence Cipher, Row Transposition Cipher.	K2	CO1
19	What are transposition techniques? Transposition techniques encrypt data by rearranging the positions of plaintext characters without changing the characters themselves. Examples: Rail Fence Cipher and Row Transposition Cipher.	K1	CO1
20	Define steganography. [Apr/May 2024] Steganography is the art and science of hiding secret information inside another medium such as an image, audio, video, or text so that the existence of the message remains concealed.	K1	CO1

Thirteen Marks Questions

S. No	Questions	Knowledge Level	Course Outcome
1	Explain about OSI security architecture model with suitable example. [Apr/May 2024] Introduction to OSI Security Architecture (ITU-T X.800) Objectives of network security Security Attacks Passive attacks Release of message contents Traffic analysis Active attacks Masquerade Replay Modification of messages	K1	CO1

S. No	Questions	Knowledge Level	Course Outcome
	Denial of Service Security Services Authentication Access Control Data Confidentiality Data Integrity Non-repudiation Availability Security Mechanisms Encryption Digital Signature Authentication Exchange Access Control Traffic Padding Routing Control Notarization Draw OSI Security Architecture diagram Suitable example (Online Banking / Internet Banking)		
2	Explain Hill cipher, Show your calculation and the result. Definition of Hill Cipher Matrix-based substitution cipher Algorithm Choose key matrix Convert plaintext into numbers (A=0 to Z=25) Divide plaintext into blocks Matrix multiplication Modulo-26 operation Show complete encryption calculation Ciphertext generation Decryption process Inverse matrix calculation Example with calculation Advantages Limitations	K2	CO1
3	Using play fair cipher algorithm encrypt the plaintext = ATTACK POSTPONED using the key = EMERGENCY. [Nov/Dec 2023] Introduction to Playfair Cipher Rules of Playfair Construct 5×5 key matrix using keyword EMERGENCY Remove duplicate letters Fill remaining alphabets (I/J together) Prepare plaintext Remove spaces Divide into digraphs Insert X wherever required Apply Playfair rules Same row Same column Rectangle rule Obtain ciphertext Final encrypted text	K3	CO1

S. No	Questions	Knowledge Level	Course Outcome
	Advantages Limitations		
4	Discuss on the attack surfaces, attack trees and the model for network security. [Nov/Dec 2024] Introduction Definition of Attack Surface Types Digital attack surface Physical attack surface Human attack surface Attack Tree Root node Child nodes Leaf nodes AND/OR relationships Draw Attack Tree Standard Network Security Model Sender Receiver Encryption Decryption Secret Key Trusted Third Party Draw Network Security Model Passive vs Active attacks Real-time example Conclusion	K1	CO1
5	Describe the various classical encryption techniques with illustrations. [Nov/Dec 2024] Introduction Classical Encryption Techniques Substitution Techniques Caesar Cipher Monoalphabetic Cipher Playfair Cipher Hill Cipher Vigenère Cipher One-Time Pad Transposition Techniques Rail Fence Cipher Row Transposition Cipher Illustrate each method with one example Advantages Limitations Comparison of substitution and transposition	K2	CO1
6	Perform encryption and decryption using Hill cipher for the following: Message PEN and key ACTIVATED [Nov/Dec 2021] Introduction Form key matrix from keyword Convert plaintext PEN into numeric values Matrix multiplication Apply modulo-26 Generate ciphertext Find inverse key matrix Perform decryption	K3	CO1

S. No	Questions	Knowledge Level	Course Outcome
	Recover original plaintext Verify result		
7	Discuss examples from real life, where the following security objectives are needed : i) Authentication. (5) ii) Access control. (5) iii) Confidentiality. (5) Suggest suitable security mechanisms to achieve them. Authentication Definition Example: Internet Banking, ATM, Gmail Login Mechanisms Password OTP Biometrics Digital Certificate Access Control Definition Example: College ERP, Hospital Records Mechanisms RBAC ACL Firewalls Authentication Confidentiality Definition Example: Online Banking, Military Communication Mechanisms AES RSA VPN SSL/TLS	K4	CO1
8	Discuss in detail about Playfair method with suitable example. Introduction History Playfair Algorithm Construction of 5×5 matrix Rules Same row Same column Rectangle rule Handling repeated letters Padding using X Encryption steps Decryption steps Neat example Advantages Disadvantages Applications	K2	CO1
9	Encrypt the following using play fair cipher using the keyword MONARCHY. “BALLOON”. Use X for blank spaces.	K3	CO1

S. No	Questions	Knowledge Level	Course Outcome
	Write keyword MONARCHY Construct 5×5 matrix Fill remaining letters Prepare plaintext BA LX LO ON Apply Playfair rules Obtain ciphertext Show complete encryption table Final ciphertext		
10	Write note on different types of security attacks and services in detail. Introduction Security Attacks Passive Attacks Release of message contents Traffic analysis Active Attacks Masquerade Replay Modification Denial of Service Comparison table (Passive vs Active) Security Services Authentication Access Control Confidentiality Integrity Non-repudiation Availability Relationship between attacks and services Real-time examples	K2	CO1

Fifteen Marks Questions

S. No	Questions	Knowledge Level	Course Outcome
1	Show your calculation and the result to encrypt the message “meet me at the usual place at ten rather eight o clock ” using Hill cipher with key $\begin{bmatrix} 7 & 3 \\ 2 & 5 \end{bmatrix}$ [Apr/May 2024] Introduction to Hill Cipher Explain matrix-based encryption technique Given key matrix: Remove spaces and convert plaintext to uppercase Convert letters into numerical values (A=0, B=1, ..., Z=25) Divide plaintext into 2-letter blocks Convert each block into a column matrix Perform matrix multiplication: $C = K \times P \pmod{26}$ Show complete calculations for each block	K3	CO1

S. No	Questions	Knowledge Level	Course Outcome
	Obtain ciphertext Mention padding (X) if plaintext length is odd Explain decryption using inverse key matrix Advantages of Hill Cipher		
2	Discuss examples from real life, where the following security objectives are needed : i) Confidentiality. (5) ii) Integrity. (5) iii) Non-repudiation. (5) Suggest suitable security mechanisms to achieve them. [Nov/Dec 2021] (i) Confidentiality Definition Real-time Examples: Internet Banking Military Communication Medical Records Cloud Storage Security Mechanisms: AES RSA SSL/TLS VPN Access Control (ii) Integrity Definition Examples: Online Examination Banking Transactions Software Downloads Security Mechanisms: SHA HMAC Digital Signature Checksums Message Authentication Code (MAC) (iii) Non-repudiation Definition Examples: Online Contracts E-commerce Digital Payments Email Transactions Security Mechanisms: Digital Signature PKI Timestamping Digital Certificates	K2	CO1
3	Discuss the rules to be followed in Playfair method. Encrypt the word “Networksecurity” with	K3	CO1

S. No	Questions	Knowledge Level	Course Outcome
	the keyword “cypto” using Playfair method.[Nov/Dec 2021] Introduction to Playfair Cipher Features of Playfair Cipher Construction of 5×5 key matrix using keyword CRYPTO Remove duplicate letters Merge I/J Fill remaining alphabets Prepare plaintext Remove spaces Convert to uppercase Divide into digraphs Insert X between repeated letters Add X if the final block is incomplete Explain Playfair Rules: Same Row Rule Same Column Rule Rectangle Rule Perform encryption step-by-step Generate ciphertext Mention decryption procedure Advantages Disadvantages		
4	A company's confidential data was intercepted and altered during transmission over an unsecured network. Using your knowledge of OSI Security Architecture, identify the type of attack and suggest suitable security services and mechanisms to prevent such incidents. Security Services Required Data Confidentiality Data Integrity Authentication Non-repudiation Access Control Suitable Security Mechanisms Encryption (AES/RSA) Digital Signature Message Authentication Code (MAC) Hash Functions (SHA) SSL/TLS Firewalls Intrusion Detection System (IDS) VPN Explain Prevention Encrypt transmitted data Verify integrity using hash/MAC Authenticate sender and receiver Secure communication channels	K4	CO1
5	An organization is building a network security model to protect against various cyberattacks. Describe the standard network security model, apply it to a secure system design, and explain	K4	CO1

S. No	Questions	Knowledge Level	Course Outcome
	how it helps defend against both passive and active attacks. Secure System Design Encryption (AES) Public Key Cryptography (RSA) Authentication Firewalls IDS/IPS VPN Digital Certificates Access Control Protection Against Passive Attacks Encryption VPN SSL/TLS Protection Against Active Attacks Digital Signatures Hash Functions MAC Authentication IDS Firewalls		

UNIT II SYMMETRIC CIPHERS

9

Number theory – Algebraic Structures – Modular Arithmetic - Euclid's algorithm – Congruence and matrices – Group, Rings, Fields, Finite Fields SYMMETRIC KEY CIPHERS: SDES – Block Ciphers – DES, Strength of DES – Differential and linear cryptanalysis – Block cipher design principles – Block cipher mode of operation – Evaluation criteria for AES – Pseudorandom Number Generators – RC4 – Key distribution.

CO2: Apply the different cryptographic operations of symmetric cryptographic algorithms

Two Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Find GCD (1970, 1066) using Euclid's algorithm. [Apr/May 2024] $1970 = 1066 \times 1 + 904$ $1066 = 904 \times 1 + 162$ $904 = 162 \times 5 + 94$ $162 = 94 \times 1 + 68$ $94 = 68 \times 1 + 26$ $68 = 26 \times 2 + 16$ $26 = 16 \times 1 + 10$ $16 = 10 \times 1 + 6$ $10 = 6 \times 1 + 4$ $6 = 4 \times 1 + 2$ $4 = 2 \times 2 + 0$ $\therefore \text{GCD}(1970, 1066) = 2$	K2	CO2
2	Compare block cipher and stream cipher. [Apr/May 2024]	K2	CO2

S. No	Question	Knowledge Level	Course Outcome
	Block Cipher Stream Cipher ----- ----- Encrypts fixed-size blocks of data. Encrypts one bit/byte at a time. More secure for bulk data. Faster for continuous data streams. Requires padding. No padding required. Example: DES, AES Example: RC4		
3	In AES how the encryption key is expanded to produce keys for the 10 rounds. [Nov/Dec 2023] AES-128 uses the Key Expansion (Key Schedule) algorithm to generate 11 round keys (one initial key and 10 round keys). It uses RotWord(), SubWord(), Rcon, and XOR operations to derive each round key.	K2	CO2
4	What are the properties hold by a good random number generator. [Nov/Dec 2023] Uniform distribution Unpredictability Independence Long period Statistical randomness	K1	CO2
5	Compare differential and linear cryptanalysis. [Nov/Dec 2024] Differential Cryptanalysis Linear Cryptanalysis ----- ----- Uses plaintext differences. Uses linear approximations. Chosen plaintext attack. Known plaintext attack. Studies input-output differences. Finds linear relations between bits.	K2	CO2
6	Calculate gcd(568, 8) using Euclidian algorithm. [Nov/Dec 2024] $568 = 8 \times 71 + 0$ $\therefore \text{GCD}(568, 8) = 8$	K2	CO2
7	Give the 5 modes of operations of block cipher. ECB (Electronic Code Book) CBC (Cipher Block Chaining) CFB (Cipher Feedback) OFB (Output Feedback) CTR (Counter Mode)	K1	CO2
8	List the parameters for the 3 AES version AES Version Key Size Block Size Rounds ----- ----- ----- ----- AES-128 128 bits 128 bits 10 AES-192 192 bits 128 bits 12 AES-256 256 bits 128 bits 14	K2	CO2
9	Compare DES and AES DES AES 56-bit key 128/192/256-bit key 64-bit block 128-bit block 16 rounds 10/12/14 rounds Less secure Highly secure	K2	CO2
10	Define finite field. A finite field (Galois Field) is a field containing a finite number of elements in which addition, subtraction, multiplication, and division (except by zero) are defined and satisfy field properties.	K1	CO2
11	What is the disadvantage of double DES? Double DES is vulnerable to the Meet-in-the-Middle attack, reducing its effective security while increasing encryption time.	K1	CO2

S. No	Question	Knowledge Level	Course Outcome
12	What is avalanche effect? The avalanche effect is a property in which a small change in the plaintext or key causes a significant change in the ciphertext, improving security.	K2	CO2
13	Write notes on RC4. RC4 is a stream cipher developed by Ron Rivest. It uses a variable-length key (40–256 bits) and generates a pseudorandom keystream for encryption using XOR operation.	K1	CO2
14	Does the set of residue classes (mod 3) form a group? Yes. The residue classes {0,1,2} under addition modulo 3 form an Abelian group, since they satisfy closure, associativity, identity, inverse, and commutativity.	K2	CO2
15	Write down the purposes of the S-box in DES Provides substitution. Introduces non-linearity. Increases confusion. Protects against cryptanalysis.	K1	CO2
16	Define confusion and diffusion Confusion: Makes the relationship between the key and ciphertext complex. Diffusion: Spreads the influence of one plaintext bit over many ciphertext bits.	K1	CO2
17	List the 4 different stages of AES. SubBytes ShiftRows MixColumns AddRoundKey	K1	CO2
18	Why modular arithmetic has been used in cryptography? Modular arithmetic enables efficient encryption and decryption, supports finite field operations, and forms the mathematical basis for algorithms such as RSA, AES, and ECC.	K2	CO2
19	List the applications of RC4. SSL/TLS (older versions) WEP WPA (older wireless security) Secure file encryption Secure communication protocols	K1	CO2
20	Why random numbers are use in network security? Random numbers are used to generate cryptographic keys, initialization vectors (IVs), session keys, nonces, passwords, and digital signatures, thereby improving the security of communication systems.	K2	CO2

Thirteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Explain the overall structure and round structure of DES with neat sketch. [Apr/May 2024] Draw Diagram Plaintext (64 bits) Initial Permutation (IP) L0 R0 	K2	CO2

S. No	Question	Knowledge Level	Course Outcome
	----- Round 1 Round 2 ... Round 16 ----- Swap Inverse IP Cipher Text Explain each stage Initial Permutation Splitting into Left & Right halves 16 Feistel rounds Final Swap Inverse Initial Permutation Round Structure Diagram Li -----> Ri+1 Ri --> Expansion(E) XOR with Key Ki S-Boxes P-Permutation XOR with Li Ri+1 Explain Expansion Permutation XOR S-box substitution P permutation Feistel function Formula $Li+1 = Ri$ $Ri+1 = Li \text{ XOR } F(Ri, Ki)$ Key Schedule Mention PC-1 Left Shift PC-2 Generates 16 keys		

S. No	Question	Knowledge Level	Course Outcome																																										
2	(i) Solve the following using RC4 stream cipher algorithm. Assume the state vector is 8 bits. (8 Marks) Key [1 0 0 2] Plaintext [6 1 5 4] (ii) Illustrate any pseudo random number generator algorithm. (5Marks) [Apr/May 2024] (i) RC4 Numerical Write headings Step 1 Initialize S array S=[0 1 2 3 4 5 6 7] Step 2 Key Scheduling Algorithm (KSA) Use $j=(j+S[i]+Key[i \bmod keylength]) \bmod 8$ Draw KSA table <table><tr><td>i</td><td>j</td><td>Swap</td></tr><tr><td>0</td><td>7</td><td>0 7</td></tr><tr><td>1</td><td>6</td><td>1 6</td></tr><tr><td>2</td><td>5</td><td>2 5</td></tr><tr><td>3</td><td>4</td><td>3 4</td></tr><tr><td>4</td><td>3</td><td>4 3</td></tr><tr><td>5</td><td>2</td><td>5 2</td></tr><tr><td>6</td><td>1</td><td>6 1</td></tr><tr><td>7</td><td>0</td><td>7 0</td></tr></table> Step 3 Pseudo Random Generation Algorithm i=i+1 j=j+S[i] Swap t=(S[i]+S[j]) mod 8 Key=S[t] Generate keystream. Step 4 Encryption Cipher = Plain XOR Key Stream Make table <table><tr><td>Plain</td><td>Key</td><td>Cipher</td></tr><tr><td>6</td><td>1</td><td>7</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>5</td><td>0</td><td>5</td></tr><tr><td>4</td><td>2</td><td>6</td></tr></table> Finally write Cipher text. (ii) Pseudo Random Number Generator Explain Linear Congruential Generator Formula	i	j	Swap	0	7	0 7	1	6	1 6	2	5	2 5	3	4	3 4	4	3	4 3	5	2	5 2	6	1	6 1	7	0	7 0	Plain	Key	Cipher	6	1	7	1	0	1	5	0	5	4	2	6	K3	CO2
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S. No	Question	Knowledge Level	Course Outcome
	$X_{i+1} = (aX_i + c) \bmod m$ Diagram Seed LCG Formula Random Number Next Seed Mention Seed Multiplier Increment Modulus Applications		
3	Illustrate the core principles of Euclid's and extended Euclid's algorithm and explain its functionality. [Nov/Dec 2024] Introduction Purpose Find GCD Extended Euclid also finds multiplicative inverse Euclid Algorithm Flow GCD(a,b) ↓ $r = a \bmod b$ ↓ $a = b$ ↓ $b = r$ ↓ Repeat until $r = 0$ Algorithm while $b \neq 0$ $r = a \bmod b$	K3	CO2

S. No	Question	Knowledge Level	Course Outcome
	$a=b$ $b=r$ return a Extended Euclid Table $ r q x y $ Explain $ax+by=\gcd(a,b)$ Applications RSA Modular inverse ECC Difference table Euclid Extended Euclid		
4	Describe detailed structure of AES and reveal the significance of transformation function. [Nov/Dec 2024] tepwise division $98=56 \times 1 + 42$ $56=42 \times 1 + 14$ $42=14 \times 3 + 0$ Therefore GCD=14 Back substitution $14=56-42$ $42=98-56$ $14=56-(98-56)$ $14=2 \times 56 - 98$ Hence $x=-1$ $y=2$ Verification	K2	CO2

S. No	Question	Knowledge Level	Course Outcome
	$98(-1)+56(2)=14$ Write algorithm also.		
5	Explain block cipher modes of operation with neat sketch. [Nov/Dec 2023] Symmetric cipher Block size =128 bits Key sizes 128 192 256 Rounds 10 12 14 Diagram Plain Text ↓ Add Round Key ↓ SubBytes ↓ ShiftRows ↓ MixColumns ↓ AddRoundKey ↓ Repeat ↓ Final Round (No MixColumns)	K1	CO2

S. No	Question	Knowledge Level	Course Outcome
	↓ Cipher Text Explain SubBytes S-box substitution ShiftRows Circular left shift MixColumns Matrix multiplication AddRoundKey XOR with key Key Expansion Explain briefly		
6	Describe Triple DES and its applications in detail. ECB $P1 \rightarrow \text{Encrypt} \rightarrow C1$ $P2 \rightarrow \text{Encrypt} \rightarrow C2$ Independent blocks. Advantages Simple Disadvantage Pattern visible. CBC $P1 \text{ XOR } IV$ ↓ Encrypt ↓ $C1$ ↓ $P2 \text{ XOR } C1$ ↓ Encrypt Advantages	K1	CO2

S. No	Question	Knowledge Level	Course Outcome
	More secure. CFB Cipher feedback. Diagram IV ↓ Encrypt ↓ XOR Plain ↓ Cipher OFB Output feedback. Explain keystream generation. CTR Counter mode. Counter ↓ Encrypt ↓ Keystream ↓ XOR Plain		
7	Explain the key generation, encryption and decryption of SDES algorithm in detail. DES Uses DES three times. Diagram Encryption Plain	K1	CO2

S. No	Question	Knowledge Level	Course Outcome
	↓ Encrypt(K1) ↓ Decrypt(K2) ↓ Encrypt(K3) ↓ Cipher Write $C = EK_3(DK_2(EK_1(P)))$ Decryption Reverse order. Key options 2-key 3-key Advantages More secure than DES. Applications ATM Banking Financial transactions		
8	Explain in detail about the entities in the symmetric cipher model with their requirements for secure usage of the model Simplified DES Block 8 bits Key 10 bits Key Generation	K3	CO2

S. No	Question	Knowledge Level	Course Outcome
	Diagram 10-bit key ↓ P10 ↓ LS-1 ↓ P8 ↓ K1 ↓ LS-2 ↓ P8 ↓ K2 Encryption IP ↓ $F_k(K1)$ ↓ Swap ↓ $F_k(K2)$ ↓ IP^{-1} Explain		

S. No	Question	Knowledge Level	Course Outcome
	Expansion XOR S-box P4 Decryption Same process Keys used		
9	Solve GCD (98,56) using extended Euclidean algorithm. Also, write the algorithm Plaintext ↓ Encryption ↓ Ciphertext ↓ Transmission ↓ Decryption ↓ Plaintext Entities Plaintext Encryption Algorithm Secret Key Ciphertext Decryption Algorithm Sender Receiver Attacker Requirements Strong algorithm Secret key confidentiality Proper key distribution Authentication Integrity	K3	CO2

S. No	Question	Knowledge Level	Course Outcome
	Applications Secure communication Banking		
10	Discuss the properties that are to be satisfied by groups, rings and fields Group Properties Closure Associativity Identity Inverse Example $(\mathbb{Z}, +)$ Abelian Group Additional property $a+b=b+a$ Ring Properties Addition forms Abelian Group Multiplication associative Distributive law Example Integers Field Properties Ring Multiplicative inverse exists Commutative multiplication Example $GF(2)$ Applications AES ECC RSA	K4	CO2

Fifteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Explain Euclid's algorithm with a step-by-step procedure. Apply it to find the GCD of two large numbers. How can this be used in key generation for cryptographic applications?	K3	CO2
2	Discuss modular arithmetic in detail. Solve: a) $275 \bmod 13$ b) Solve $5x \equiv 1 \pmod{17}$ c) Find all integers x such that $3x \equiv 6 \pmod{9}$	K3	CO2
3	Define and differentiate Groups, Rings, and Fields with suitable examples. Why are fields particularly important in cryptography?	K4	CO2
4	Matrices are used in Hill Cipher encryption. Explain how matrix arithmetic and modular operations are applied in cryptography, and show an example using a 2×22 matrix for encryption and decryption.	K4	CO2
5	What are finite fields? Explain the structure of $GF(2^n)$ and its application in AES with examples.	K3	CO2

UNIT III ASYMMETRIC CRYPTOGRAPHY

9

MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY: Primes – Primality Testing – Factorization – Euler's totient function, Fermat's and Euler's Theorem – Chinese Remainder Theorem – Exponentiation and logarithm ASYMMETRIC KEY CIPHERS: RSA cryptosystem – Key distribution – Key management – Diffie Hellman key exchange – Elliptic curve arithmetic – Elliptic curve cryptography.

CO3: Apply the different cryptographic operations of public key cryptography

Two Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Find the remainder when 7^{23} is divided by 11 using Fermat's Little Theorem. [Apr/May 2024] By Fermat's Little Theorem, $7^{10} \equiv 1 \pmod{11}$ $7^{23} = 7^{20} \times 7^3 = (7^{10})^2 \times 7^3 \equiv 1^2 \times 7^3 \pmod{11} \equiv 7^3 \pmod{11}$ $7^3 = 343 \pmod{11} \equiv 343 \div 11 = 31 \text{ remainder } 2$ $\therefore$ Remainder = 2	K1	CO3
2	How does the Chinese remainder theorem contribute to the efficient modular arithmetic in encryption and decryption process? [Apr/May 2024] The Chinese Remainder Theorem (CRT) reduces large modular computations into smaller independent computations, thereby speeding up encryption and decryption. It is widely used in RSA to improve decryption efficiency.	K1	CO3
3	Does Fermat's theorem hold for $p=5$ and $a=2$? Justify. [Nov/Dec 2023] Fermat's theorem states that if p is prime and a is not divisible by p , $a^{p-1} \equiv 1 \pmod{p}$ Here, $2^{5-1} = 2^4 = 16 \equiv 1 \pmod{5}$ Since $16 \bmod 5 = 1$, Fermat's theorem holds	K2	CO3
4	What is message authentication code? How it differ from hash function? [Nov/Dec 2024] A Message Authentication Code (MAC) is a cryptographic value generated using a secret key to	K1	CO3

S. No	Question	Knowledge Level	Course Outcome
	verify message integrity and authenticity.		
5	Find $8^{-1} \pmod{173}$. [Nov/Dec 2024] Find xxx such that $8x \equiv 1 \pmod{17}$ Since $8 \times 15 = 120 \equiv 1 \pmod{17}$ $\therefore 8^{-1} \equiv 15 \pmod{17}$	K2	CO3
6	Mention the advantages of elliptic curve cryptography. [Nov/Dec 2024] Smaller key size Faster computation Lower memory usage Higher security Suitable for mobile and IoT devices	K1	CO3
7	What is public key cryptography? Public Key Cryptography is an asymmetric encryption technique that uses two keys: a public key for encryption and a private key for decryption, ensuring secure communication.	K1	CO3
8	Compare symmetric key cryptography and public key cryptography. Symmetric Key Public Key Uses one secret key Uses public and private keys Faster Slower Suitable for bulk encryption Suitable for key exchange and digital signatures Example: AES Example: RSA	K2	CO3
9	17 as the two primes. Find $n = 11$ and $q = 17$. Consider the RSA encryption method with $p(n) \cdot \phi(n)$ $p=11, q=17$ $n=pq=11 \times 17=187$ $\phi(n)=(11-1)(17-1)=10 \times 16=160$ $\therefore n=187, \phi(n)=160$	K2	CO3
10	Define primitive root A primitive root of a prime number p is an integer whose powers generate all non-zero integers modulo p .	K1	CO3
11	State Fermat's theorem If p is a prime number and a is not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$	K1	CO3
12	Name any 2 methods for testing prime numbers. Fermat Primality Test Miller-Rabin Primality Test	K1	CO3
13	Define Euler's totient function. Euler's Totient Function, denoted by $\phi(n)$, is the number of positive integers less than n that are relatively prime to n .	K1	CO3
14	State Euler's theorem. If a and n are relatively prime, then $a^{\phi(n)} \equiv 1 \pmod{n}$	K1	CO3
15	$\phi(35)$ and $\phi(37)$ Determine. $35=5 \times 7$ $\phi(35)=35(1-\frac{1}{5})(1-\frac{1}{7})=24$ $\phi(37)=37-1=36$ $\therefore \phi(35)=24, \phi(37)=36$	K2	CO3
16	List the 6 ingredients of public key encryption.	K1	CO3

S. No	Question	Knowledge Level	Course Outcome
	Plaintext Encryption Algorithm Public Key Private Key Ciphertext Decryption Algorithm		
17	List the 5 possible approaches to attacking the RSA algorithm. Brute-force attack Mathematical attack (Factoring n) Timing attack Chosen ciphertext attack Chosen plaintext attack	K1	CO3
18	Define discrete logarithm. The discrete logarithm problem is to find an integer xxx such that $ax \equiv b \pmod{p}$ $a^x \equiv b \pmod{p} \Leftrightarrow x \equiv \log_a b \pmod{p}$ It forms the basis of the security of Diffie–Hellman and ECC.	K1	CO3
19	What is the principal attraction of ECC, compared to RSA? [Nov/Dec 2021] The principal attraction of ECC is that it provides the same level of security as RSA using much smaller key sizes, resulting in faster computation, reduced memory usage, and lower power consumption.	K1	CO3
20	Perform encryption for the plaintext M=88 using the RSA algorithm. P=17, q=11 and public component e=7 $p=17, q=11, e=7, M=88$ $n=pq=17 \times 11=187, n=pq=17 \times 11=187, n=pq=17 \times 11=187$ RSA Encryption: $C = M^e \pmod{n}$ $C = 88^7 \pmod{187}$ Using modular exponentiation, $C = 11$ ∴ Ciphertext = 11	K2	CO3

Thirteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Solve the RSA algorithm with p=13 and q=17. [Apr/May 2024] Step 1: Select Prime Numbers $p = 13$ $q = 17$ Step 2: Calculate n $n = p \times q$ $= 13 \times 17$ $= 221$ Step 3: Euler Totient $\phi(n)$ $= (13-1)(17-1)$ $= 12 \times 16$	K3	CO3

S. No	Question	Knowledge Level	Course Outcome
	$n=192$ Step 4: Select Public Key (e) Choose $1 < e < 192$ $\gcd(e, 192) = 1$ Example $e=5$ Step 5: Find Private Key Find d such that $de \equiv 1 \pmod{192}$ Result $d=77$ $(5 \times 77 = 385 \equiv 1 \pmod{192})$ Public Key (5,221) Private Key (77,221) Encryption Formula $C = M^e \pmod{n}$ Example Take $M=10$ Then $C = 10^5 \pmod{221}$ Decryption Formula $M = C^d \pmod{221}$		
2	Solve Diffie – Hellman scheme with $q=11$, primitive root $a=2$, $Y_a=9$ and $Y_b=3$. [Apr/May 2024] Step 1 Find Alice's private key	K3	CO3

S. No	Question	Knowledge Level	Course Outcome
	Need $2^{XA} \bmod 11 = 9$ Check $2^6 = 64$ $64 \bmod 11 = 9$ $XA = 6$ Step 2 Find Bob's private key $2^{XB} \bmod 11 = 3$ Check $2^8 = 256$ $256 \bmod 11 = 3$ $XB = 8$ Step 3 Alice computes $K = (YB)^{XA} \bmod 11$ $= 3^6 \bmod 11$ $= 3$ Step 4 Bob computes $K = (YA)^{XB} \bmod 11$ $= 9^8 \bmod 11$ $= 3$ Shared Secret $K = 3$		
3	Determine the plaintext from the cipher text $C=106$, $N=143$, $e=11$ using RSA. [Nov/Dec 2023] Step 1 Factor $143 = 11 \times 13$ Therefore	K3	CO3

S. No	Question	Knowledge Level	Course Outcome
	p=11 q=13 Step 2 Find Totient $\phi=(10 \times 12)=120$ Step 3 Find d Need $11d \equiv 1 \pmod{120}$ Result d=11 $(11 \times 11=121 \equiv 1 \pmod{120})$ Step 4 Decrypt Formula $M=C^d \pmod{143}$ $=106^{11} \pmod{143}$		
4	Assume the client C (sender) wants to communicate with client C1 (Receiver) using Diffie – Hellman scheme. How can it be achieved? Explain with numerical example. [Nov/Dec 2023]	K4	CO3
5	Illustrate the core ideology behind Fermat's and Euler's theorem and explain its applications. Give an example for each. [Nov/Dec 2024] If p is prime $\gcd(a,p)=1$ Then $a^{(p-1)} \equiv 1 \pmod{p}$ If $\gcd(a,n)=1$ then $a^{\phi(n)} \equiv 1 \pmod{n}$	K2	CO3
6	Explain in detail about ECC algorithm with a an example. Elliptic Curve Equation $y^2=x^3+ax+b$ Condition	K2	CO3

S. No	Question	Knowledge Level	Course Outcome
	$4a^3+27b^2 \neq 0$ Diagram Draw Elliptic Curve (Exam sketch) Key Generation Private key d Public key $Q=dP$ where P =Base Point Encryption Cipher= $(kP, Pm+kQ)$ Decryption Recover Pm		
7	Explain public key cryptography and when it is preferred? Diagram Public Key ↓ Encryption ↓ Cipher Text ↓ Private Key ↓ Decryption Explain Key generation Encryption Decryption	K1	CO3

S. No	Question	Knowledge Level	Course Outcome
	Examples RSA ECC ElGamal Preferred Internet Digital Signature Key Exchange Email Security Advantages No shared secret Better scalability		
8	State and Prove Fermat's theorem and Euler's theorem	K2	CO3
9	State and prove Chinese remainder theorem. What are the last two digits of 4919? CRT Statement If $m_1, m_2, \dots, m_n$ are pairwise coprime Then Unique solution exists modulo $M = m_1 m_2 \dots m_n$ Proof Explain Product M M_i Modular inverse Construction formula Last Two Digits Need $49^{19} \bmod 100$ Observe $49^2 = 2401$ $\equiv 1 \pmod{100}$ Therefore 49^{19} $= 49(49^2)^9$	K2	CO3

S. No	Question	Knowledge Level	Course Outcome
	$\equiv 49 \times 1$ $\equiv 49$ Answer 49		
10	State Chinese Remainder Theorem and find X for the given set of congruent equations using CRT $X \equiv 1 \pmod{5}$ $X \equiv 2 \pmod{7}$ $X \equiv 3 \pmod{9}$ $X \equiv 4 \pmod{11}$	K3	CO3

Fifteen Mark Questions

S. No	Question	Knowledge Level	Course Outcome
1	(Alice and Bob exchange a shift cipher key using the Diffie-Hellman key exchange. They agree to use the prime $p = 11$ for their cyclic group $\mathbb{Z}_{11}$ and $g = 7$ as the generator. (i) Assume Alice uses the secret value $a = 6$ and Bob the secret value $b = 8$. Compute the public values and the final key that Alice and Bob exchange. (ii) Assume Alice and Bob exchange the values $X = g^a = 5 \pmod{11}$ and $Y = g^b = 10 \pmod{11}$. Determine the secret key they both exchanged! Describe your computation steps to determine the key. [Nov/Dec 2023]	K3	CO3
2	The ECC cryptosystem defined by $y^2 = x^3 + x + 6$ over F_{11} , and $G = (2, 7)$. B's secret key is 3. A wishes to encrypt the message $P_m = (10, 9)$ and chooses the random value $k = 2$. Find B's public key and determine the cipher text C_m . [Nov/Dec 2023]	K3	CO3
3	Explain the RSA algorithm with an example. Given $p = 13$, $q = 17$: • Compute n and $\phi(n)$. • Choose $e = 5$ and find the private key d. • Encrypt and decrypt the message $M = 9$	K3	CO3
4	State and prove the Chinese Remainder Theorem (CRT). Use CRT to solve the following system of congruences: $x \equiv 2 \pmod{3}$ $x \equiv 3 \pmod{4}$ $x \equiv 1 \pmod{5}$	K3	CO3
5	Define Euler's totient function $\phi(n)$. Compute $\phi(n)$ for $n = 45$ and $n = 77$, and verify Euler's theorem for a number $a = 4$ modulo 45. Also, state and prove Euler's theorem.	K3	CO3

UNIT IV INTEGRITY AND AUTHENTICATION ALGORITHMS

9

Authentication requirement – Authentication function – MAC – Hash function – Security of hash function: HMAC, CMAC – SHA – Digital signature and authentication protocols – DSS – Schnorr Digital Signature Scheme – ElGamal cryptosystem – Entity Authentication: Biometrics, Passwords, Challenge Response protocols – Authentication applications – Kerberos MUTUAL TRUST: Key management and distribution – Symmetric key distribution using symmetric and asymmetric encryption – Distribution of public keys – X.509 Certificates.

CO4: Apply the various Authentication schemes to simulate different applications

Two Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Define digital signature. What are the properties of digital signatures? [Apr/May 2024] Authentication Integrity Non-repudiation Unforgeability Verifiability	K1	CO4
2	Why is mutual trust important in network authentication? Mutual trust ensures that both communicating parties verify each other's identity, preventing impersonation, replay attacks, and unauthorized access while establishing secure communication.	K1	CO4
3	Why do we need hash function? [Nov/Dec 2024] Verify data integrity. Detect message modifications. Generate digital signatures. Support password storage and message authentication.	K1	CO4
4	What is authentication header (AH) in IPsec? The Authentication Header (AH) is an IPsec protocol that provides data integrity, data origin authentication, and protection against replay attacks, but it does not provide confidentiality.	K1	CO4
5	Compare symmetric and asymmetric key distribution. Symmetric Key Distribution Asymmetric Key Distribution Uses one shared secret key. Uses public and private keys. Faster but difficult key distribution. Easier and more secure key exchange. Example: KDC Example: RSA	K1	CO4
6	What is the role of key distribution center (KDC)? A Key Distribution Center (KDC) is a trusted third party that securely generates, distributes, and manages secret session keys between communicating users.	K1	CO4
7	Write the challenges in key distribution. [Nov/Dec 2024] Secure key exchange Key management Key storage Key revocation Scalability in large networks	K2	CO4
8	What is the significance of authentication protocols? Authentication protocols verify the identity of communicating parties, prevent unauthorized access, and protect against replay, impersonation, and man-in-the-middle attacks.	K1	CO4
9	Write the schemes for the distribution of public keys. [Nov/Dec 2023] Public Announcement Publicly Available Directory Public Key Authority Public Key Certificates	K1	CO4
10	What is the purpose of HMAC? HMAC (Hash-based Message Authentication Code) is used to verify the integrity and authenticity of a message by combining a cryptographic hash function with a secret key.	K1	CO4
11	Define entity authentication. Entity Authentication is the process of verifying the identity of a communicating user or system before granting access to network resources.	K1	CO4
12	Compare MAC and digital signature. MAC Digital Signature	K1	CO4

S. No	Question	Knowledge Level	Course Outcome
	Uses a shared secret key. Uses public/private key pair. Provides authentication and integrity. Provides authentication, integrity, and non-repudiation. Faster. Computationally more expensive.		
13	What are the requirements of authentication? Entity authentication Data origin authentication Integrity Freshness (Replay protection) Confidentiality (when required)	K1	CO4
14	List the properties of hash function. One-way property Fixed-length output Collision resistance Preimage resistance Second preimage resistance Fast computation	K1	CO4
15	Write a simple authentication dialogue used in Kerberos. [Apr/May 2024] Client requests Ticket Granting Ticket (TGT) from Authentication Server (AS). AS issues TGT. Client requests service ticket from Ticket Granting Server (TGS). TGS issues service ticket. Client accesses the application server using the service ticket.	K2	CO4
16	Define message authentication code (MAC). A Message Authentication Code (MAC) is a fixed-size value generated using a secret key and a message to verify message integrity and authenticity.	K1	CO4
17	List the applications of X.509 certificate. [Nov/Dec 2023] SSL/TLS Secure e-mail (S/MIME) VPN authentication Digital signatures Public Key Infrastructure (PKI)	K2	CO4
18	List the steps of Kerberos protocol. Client login Authentication Server (AS) verification Ticket Granting Ticket (TGT) issuance Ticket Granting Server (TGS) request Service ticket issuance Secure access to the application server	K1	CO4
19	Define DSS. Digital Signature Standard (DSS) is a U.S. standard for digital signatures based on the Digital Signature Algorithm (DSA), providing authentication, integrity, and non-repudiation.	K1	CO4
20	What is ElGamal signature scheme? The ElGamal Signature Scheme is a public-key digital signature algorithm based on the discrete logarithm problem, used to authenticate digital messages and ensure integrity and non-repudiation.	K1	CO4

Thirteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Write down the examples for the schnorr digital signature standard. Provide numerical example for the same. [Nov/Dec 2023] Introduction - Schnorr Signature is a digital signature scheme based on the Discrete Logarithm Problem (DLP). - It provides: - Authentication - Integrity - Non-repudiation - It is efficient and requires shorter signatures than RSA. System Parameters Choose Prime p Prime q (q divides $p-1$) Generator g Private Key x Public Key $y = g^x \text{ mod } p$ Signing Process Choose random k Compute $r = g^k \text{ mod } p$ Compute $e = H(M \parallel r)$ Compute $s = k + xe \text{ mod } q$ Signature (e, s) Verification Receiver computes $r' = g^s y^{-e} \text{ mod } p$ Then $e' = H(M \parallel r')$ Accept if $e = e'$ Neat Diagram Message ↓ Random k ↓ $r = g^k \text{ mod } p$ ↓ Hash(M, r)	K3	CO4

S. No	Question	Knowledge Level	Course Outcome
	↓ e ↓ $s=k+xe$ ↓ Signature(e,s) ↓ Verification Numerical Example Choose $p=23$ $q=11$ $g=2$ Private key $x=4$ Public Key $y=2^4 \text{ mod } 23$ $=16$ Choose $k=5$ Compute $r=2^5 \text{ mod } 23$ $=9$ Assume $e=3$ Compute $s=(5+4 \times 3) \text{ mod } 11$ $=17 \text{ mod } 11$ $=6$ Signature (3,6) Explain verification briefly. Advantages • Short signature • Fast verification • Less storage • High security Applications • Blockchain		

S. No	Question	Knowledge Level	Course Outcome
	- Secure communication - Software authentication		
2	What are the properties of Hash? Show how the 512 bit input message blocks are converted into 160 bit message digest using SHA-1 algorithm. [Nov/Dec 2023] Introduction Hash Function converts Any length input ↓ Fixed length output SHA-1 produces 160-bit Message Digest Properties of Hash Function - Deterministic - Fixed-length output - One-way function - Collision resistance - Pre-image resistance - Second pre-image resistance - Avalanche effect - Fast computation SHA-1 Block Diagram Input Message ↓ Padding ↓ Append Length ↓ 512-bit Blocks ↓ 80 Rounds ↓ 160-bit Output Explain Each Step Step 1 Padding Append 1 000...	K3	CO4

S. No	Question	Knowledge Level	Course Outcome
	until message becomes 448 mod512 Step 2 Append 64-bit Message Length Now block becomes 512 bits Step 3 Initialize H0 H1 H2 H3 H4 Each 32 bits Step 4 Prepare 80 Words Step 5 80 Rounds Use AND OR NOT XOR and left rotations. Step 6 Produce 160-bit Digest Applications • Password hashing • Digital signatures • File integrity		
3	Explain the need for authentication function and discuss a digital signature based scheme. [Nov/Dec 2024] Introduction Authentication ensures • Sender identity • Message integrity • Non-repudiation	K3	CO4

S. No	Question	Knowledge Level	Course Outcome
	Need • Prevent impersonation • Prevent replay attack • Detect message modification • Verify sender Authentication Methods • Message Authentication Code • Hash Function • Digital Signature Digital Signature Diagram Message ↓ Hash ↓ Private Key ↓ Digital Signature ↓ Transmission ↓ Hash ↓ Public Key ↓ Verification Working Signing Hash(M) ↓ Encrypt with Private Key ↓ Signature Verification		

S. No	Question	Knowledge Level	Course Outcome
	Decrypt Signature ↓ Compare Hash Advantages • Authentication • Integrity • Non-repudiation Applications • Banking • E-commerce • Government documents		
4	Describe the performance of symmetric key distribution using asymmetric encryption. [Nov/Dec 2024] Introduction Uses • Public key to distribute • Secret key for encryption Diagram Sender ↓ Generate Session Key ↓ Encrypt Session Key ↓ Receiver Public Key ↓ Transmit ↓ Receiver Private Key ↓ Recover Session Key Steps 1. Generate session key. 2. Encrypt using receiver's public key. 3. Send encrypted session key. 4. Receiver decrypts. 5. Both communicate using symmetric encryption.	K2	CO4

S. No	Question	Knowledge Level	Course Outcome
	Advantages • Fast communication • Secure key distribution • Combines strengths of both cryptosystems Applications • SSL • TLS • HTTPS		
5	With neat diagram explain the steps involved in SHA algorithm for encrypting the message with maximum length of less than 2128 bits and produce as a output of message digest 512. [Apr/May 2024] Introduction SHA-512 belongs to SHA-2 Family Produces 512-bit Digest Supports message length $<2^{128}$ bits Diagram Message ↓ Padding ↓ Append 128-bit Length ↓ 1024-bit Blocks ↓ 80 Rounds ↓ 512-bit Digest Explain • Padding • Append Length • Initialize 8 Registers • Message Schedule • Compression Function • Final Digest Registers A B	K1	CO4

S. No	Question	Knowledge Level	Course Outcome
	C D E F G H Each 64 bits Advantages • More secure than SHA-1 • Collision resistant Applications • Blockchain • SSL • Certificates		
6	Discuss Kerberos in detail and explain how it supports mutual authentication. Introduction Kerberos is a Trusted Third Party Authentication Protocol Uses Ticket Based Authentication Components • Client • Authentication Server (AS) • Ticket Granting Server (TGS) • Service Server Diagram Client ↓ Authentication Server ↓ Ticket Granting Server ↓ Application Server Working 1. User login 2. Receive TGT 3. Request service ticket	K2	CO4

S. No	Question	Knowledge Level	Course Outcome
	4. Receive service ticket 5. Access service Mutual Authentication Server also proves identity to client. Therefore Both authenticate each other. Advantages • Single Sign-On • Mutual Authentication • Password not sent • Secure Applications • Windows Domain • Enterprise Networks		
7	Describe digital signature algorithm and show how signing and verification is done using DSS. [Apr/May 2024] Introduction DSS is based on DSA Algorithm Provides • Authentication • Integrity • Non-repudiation Key Generation Choose p q g x Public Key $y = g^x \text{ mod } p$ Signing Random k Compute $r = (g^k \text{ mod } p) \text{ mod } q$ Compute $s = k^{-1}(H(M) + xr) \text{ mod } q$ Signature (r,s) Verification Compute $w = s^{-1}$ Compute u1	K1	CO4

S. No	Question	Knowledge Level	Course Outcome
	u_2 Finally $v = ((g^{u_1} y^{u_2}) \bmod p) \bmod q$ Accept if $v = r$ Diagram Message ↓ Hash ↓ DSA Signing ↓ Signature ↓ Verification Applications - E-Governance - Banking - Digital Certificates		
8	Describe challenge-response protocol and its advantages with a real-time application. Introduction Authentication protocol based on Random Challenge Diagram Client ↓ Challenge ↓ Server ↓ Response ↓ Verification Working	K2	CO4

S. No	Question	Knowledge Level	Course Outcome
	1. Server sends random number. 2. Client encrypts challenge. 3. Sends response. 4. Server verifies. Advantages • Prevents replay attack • No password transmission • Strong authentication Real-Time Applications • ATM • Banking • Wi-Fi Authentication • VPN • Smart Cards		
9	Discuss the concept and components of X.509 certificate format. Introduction Digital certificate used in PKI Issued by Certificate Authority Diagram Certificate ↓ Version ↓ Serial Number ↓ Issuer ↓ Validity ↓ Subject ↓ Public Key ↓ Digital Signature	K2	CO4

S. No	Question	Knowledge Level	Course Outcome
	Components • Version • Serial Number • Signature Algorithm • Issuer Name • Validity • Subject • Public Key • Extensions • CA Signature Working 1. CA verifies user. 2. Generates certificate. 3. Signs certificate. 4. User presents certificate. 5. Receiver verifies CA signature. Applications • HTTPS • SSL • Email Security • VPN		
10	Explain CMAC algorithm with example and its advantages over HMAC. Introduction CMAC Cipher-based Message Authentication Code Uses AES instead of hash function. Diagram Message ↓ Split into Blocks ↓ AES Encryption ↓ Final Block ↓ MAC Working 1. Generate subkeys. 2. Divide message.	K1	CO4

S. No	Question	Knowledge Level	Course Outcome
	3. XOR with subkey. 4. AES encryption. 5. Final output is MAC.		

Fifteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Explain in detail the various digital signature schemes: DSS, ElGamal, and Schnorr.	K2	CO4
2	Explain in detail the key management and distribution methods. How does X.509 help in this?	K2	CO4
3	Design and explain an end-to-end authentication protocol using public key cryptography.	K5	CO4
4	Your work as an information technology administrator for multinational firm. Describe how you would utilize Kerberos to improve network security and simplify user authentication and authorization. What components and processes would you need to put in place to do this? [Apr/May 2024]	K4	CO4
5	Compare authentication protocols: Password-based, Biometric, and Challenge-response.	K3	CO4

UNIT V CYBER CRIMES AND CYBER SECURITY

9

Cyber Crime and Information Security – classifications of Cyber Crimes – Tools and Methods – Password Cracking, Keyloggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security

CO5: Understand various cyber crimes and cyber security.

Two Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Define cybercrime. Cybercrime is any unlawful activity in which a computer, network, or digital device is used as a tool, target, or medium to commit crimes such as hacking, identity theft, phishing, or data theft.	K1	CO5
2	How are cybercrimes classified? [Nov/Dec 2024] - Crimes against Individuals - Crimes against Property - Crimes against Organizations - Crimes against Society - Crimes against Government	K2	CO5
3	Define malware. Malware (Malicious Software) is software intentionally designed to damage, disrupt, steal data, or gain unauthorized access to computer systems. Examples include viruses, worms, Trojans, spyware, and ransomware.	K1	CO5

S. No	Question	Knowledge Level	Course Outcome
4	Write the key principles and practices of cloud security. [Apr/May 2024] • Confidentiality • Integrity • Availability • Identity and Access Management (IAM) • Data Encryption • Regular Backup and Disaster Recovery	K1	CO5
5	What are common wireless security threats? • Eavesdropping • Rogue Access Points • Man-in-the-Middle (MITM) attacks • Denial of Service (DoS) • Wi-Fi Password Cracking	K1	CO5
6	Define cybercrime and explain why it poses significant challenges to individuals, organizations and society. [Apr/May 2024] Cybercrime is any illegal activity involving computers or networks. It poses significant challenges because it causes financial loss, identity theft, data breaches, privacy violations, service disruption, and threats to national security.	K2	CO5
7	What is network access control? Network Access Control (NAC) is a security mechanism that authenticates users and devices before granting access to network resources, ensuring that only authorized users can connect.	K1	CO5
8	What is ransomware? Ransomware is a type of malware that encrypts a victim's files or locks the system and demands a ransom payment to restore access.	K1	CO5
9	What is intrusion detection system (IDS)? An Intrusion Detection System (IDS) is a security system that continuously monitors network or system activities to detect unauthorized access, attacks, or policy violations and generates alerts.	K1	CO5
10	List the layers of security in cloud computing. • Physical Security • Network Security • Host Security • Application Security • Data Security • Identity and Access Management (IAM)	K1	CO5
11	What are botnets? Botnets are networks of compromised computers (bots) controlled remotely by an attacker to perform malicious activities such as DDoS attacks, spam distribution, and malware propagation.	K1	CO5
12	What is the key logger? A Keylogger is a hardware or software tool that secretly records every keystroke typed by a user to steal passwords, banking details, and other confidential information.	K1	CO5
13	How to prevent SQL injection attack? [Nov/Dec 2023] • Using parameterized queries (Prepared Statements) • Validating user input • Using stored procedures • Escaping special characters	K1	CO5
14	Mention any two cyber laws in India.	K1	CO5

S. No	Question	Knowledge Level	Course Outcome										
	- Information Technology (IT) Act, 2000 - Information Technology (Amendment) Act, 2008												
15	What is zero-day vulnerability? - Information Technology (IT) Act, 2000 - Information Technology (Amendment) Act, 2008	K1	CO5										
16	List three preventive steps for password protection. - Use strong and unique passwords. - Enable Multi-Factor Authentication (MFA). - Change passwords regularly and never share them.	K1	CO5										
17	Differentiate between active and passive cyber attacks. <table><tr><td>Active Cyber Attack</td><td>Passive Cyber Attack</td></tr><tr><td>Modifies or disrupts data.</td><td>Only monitors or intercepts data.</td></tr><tr><td>Affects integrity and availability.</td><td>Affects confidentiality.</td></tr><tr><td>Easier to detect.</td><td>Difficult to detect.</td></tr><tr><td>Example: DoS attack</td><td>Example: Eavesdropping</td></tr></table>	Active Cyber Attack	Passive Cyber Attack	Modifies or disrupts data.	Only monitors or intercepts data.	Affects integrity and availability.	Affects confidentiality.	Easier to detect.	Difficult to detect.	Example: DoS attack	Example: Eavesdropping	K2	CO5
Active Cyber Attack	Passive Cyber Attack												
Modifies or disrupts data.	Only monitors or intercepts data.												
Affects integrity and availability.	Affects confidentiality.												
Easier to detect.	Difficult to detect.												
Example: DoS attack	Example: Eavesdropping												
18	What are the classifications of cybercrimes? [Nov/Dec 2023] - Crimes against Individuals - Crimes against Property - Crimes against Organizations - Crimes against Society - Crimes against Government	K1	CO5										
19	What is information security? Information Security is the practice of protecting information and information systems from unauthorized access, disclosure, modification, destruction, or disruption while ensuring confidentiality, integrity, and availability (CIA).	K1	CO5										
20	What is a key logger? [Nov/Dec 2024] A Keylogger is malicious software or hardware that secretly records the keystrokes typed on a keyboard to capture sensitive information such as usernames, passwords, and credit card details.	K1	CO5										

Thirteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Discuss the various security challenges in cloud computing and depict the complete ElGamal cryptosystem in detail. [Nov/Dec 2023] Introduction Cloud computing provides - On-demand services - Resource sharing - Scalability Security becomes a major concern due to shared infrastructure.	K4	CO5

S. No	Question	Knowledge Level	Course Outcome
	Cloud Security Architecture <pre> Users Internet Cloud Provider Applications Virtual Machines Storage </pre> Security Challenges • Data Breach • Data Loss • Insider Threat • Account Hijacking • Insecure APIs • Denial of Service (DoS/DDoS) • Multi-tenancy Issues • Malware Injection • Compliance & Privacy • Weak Access Control Mitigation • AES Encryption • Multi-Factor Authentication • Firewalls • IDS/IPS • Backup & Disaster Recovery • IAM Policies		
2	Explain various attacks on Kerberos and depict how it could withstand from all such violations with neat sketch. [Nov/Dec 2023] Introduction ElGamal is an asymmetric encryption algorithm based on the Discrete Logarithm Problem. Key Generation Choose Prime p Primitive root g Private key x Public key $y = g^x \text{ mod } p$ Encryption Random k Compute $C1 = g^k \text{ mod } p$ $C2 = M \times y^k \text{ mod } p$ Cipher	K2	CO5

S. No	Question	Knowledge Level	Course Outcome
	(C1,C2) Decryption $M = C2 \times (C1^x)^{-1} \text{ mod } p$ Diagram Plaintext ↓ Key Generation ↓ Encryption ↓ (C1,C2) ↓ Decryption ↓ Original Message Advantages • Secure against brute force • Supports digital signatures • Based on discrete logarithm Applications • PGP • Secure Email • Digital Signature		
3	Describe the implementation of network access control in cloud using EAP. [Nov/Dec 2024] Introduction NAC controls device access based on security policies. EAP = Extensible Authentication Protocol. Components • Client • Authenticator • Authentication Server • Cloud Resources Diagram Client ↓ Authenticator	K3	CO5

S. No	Question	Knowledge Level	Course Outcome
	↓ RADIUS Server ↓ Cloud Resource Implementation Steps 1. Device requests access. 2. EAP authentication starts. 3. User credentials verified. 4. Security posture checked. 5. Access granted/denied. 6. Continuous monitoring. Authentication Techniques • Password • Certificates • OTP • Biometrics • MFA Benefits • Unauthorized access prevention • Policy enforcement • Device validation Applications • Enterprise Cloud • Campus Networks		
4	Illustrate the services and phases of operation in IEEE 802.11i. [Nov/Dec 2024] Introduction IEEE 802.11i improves Wi-Fi security using WPA2. Services • Authentication • Confidentiality • Integrity • Key Management Phases 1. Discovery 2. Authentication 3. Key Generation 4. Protected Data Transfer 5. Termination Diagram Discovery ↓ Authentication	K1	CO5

S. No	Question	Knowledge Level	Course Outcome
	↓ Key Exchange ↓ Secure Communication ↓ Disconnect Advantages • Strong Encryption (AES) • Secure Authentication • Dynamic Keys Applications • Enterprise Wi-Fi • Campus Networks		
5	Describe how spyware functions and its impact on privacy and data security. What steps can individual and organizations take to detect and prevent any spyware infections. [Apr/May 2024] Introduction Spyware secretly collects user information. Types • Adware • Keylogger • Tracking Cookies • Browser Hijacker • Trojan Spyware Working Internet ↓ Spyware Installation ↓ Background Monitoring ↓ Collect Information ↓ Attacker Impact • Password Theft	K4	CO5

S. No	Question	Knowledge Level	Course Outcome
	- Financial Loss - Privacy Violation - Identity Theft - System Slowdown Detection - Antivirus - Anti-spyware - Firewall - Task Manager - Network Monitoring Prevention - Software Updates - Trusted Downloads - MFA - Browser Protection - User Awareness		
6	Discuss the security risks associated with public Wi-Fi networks and the best practices for ensuring wireless security on personal devices. [Apr/May 2024] Risks - Man-in-the-Middle - Rogue Access Point - Packet Sniffing - Session Hijacking - Malware Distribution Diagram User ↓ Public Wi-Fi ↓ Attacker ↓ Internet Best Practices - Use VPN - HTTPS Websites - Disable Auto Connect - Enable Firewall - Avoid Banking - MFA - Keep Devices Updated Applications - Secure Remote Work - Business Travel	K4	CO5

S. No	Question	Knowledge Level	Course Outcome
7	Explain SQL Injection and methods to prevent it. SQL Injection inserts malicious SQL into application inputs. Example SELECT * FROM users WHERE username='admin' AND password='1234'; Injected input ' OR '1'='1 Attack Flow User Input ↓ Application ↓ Database ↓ Unauthorized Access Prevention • Prepared Statements • Parameterized Queries • Input Validation • Stored Procedures • Least Privilege • WAF • Error Handling Advantages Secure database access.	K2	CO5
8	Write a short note on spyware, keyloggers and their detection. Spyware Collects information secretly. Examples • Pegasus • Browser Spyware Keylogger Records • Keystrokes • Passwords • Banking Credentials Types • Hardware • Software Detection • Antivirus	K1	CO5

S. No	Question	Knowledge Level	Course Outcome
	• Anti-malware • Process Monitoring • Registry Check • Network Monitoring Prevention • Updates • Strong Authentication • Trusted Software		
9	Describe the techniques to secure wireless networks from cyber attacks. Threats • Rogue AP • Eavesdropping • DoS • Packet Sniffing • Evil Twin • MAC Spoofing Protection • WPA3 • AES Encryption • Strong Password • Disable WPS • VPN • Firewall • IDS/IPS • Firmware Updates Diagram Users ↓ Secure Router ↓ Firewall ↓ Internet	K2	CO5
10	Explain the various cloud security issues and mitigation techniques with real time examples. Issues • Data Leakage • Malware • Insider Attack • Misconfiguration • API Vulnerability • Compliance • DoS Mitigation • Encryption	K3	CO5

S. No	Question	Knowledge Level	Course Outcome
	• IAM • MFA • Backup • Security Monitoring • SIEM • Zero Trust Real-Time Examples • AWS S3 bucket misconfiguration • Azure credential leak • Google Cloud IAM misuse		

Fifteen Marks Questions

S. No	Question	Knowledge Level	Course Outcome
1	Your company is implementing network access control (NAC) to improve network security. Outline the implementation process, including policy development, authentication techniques and enforcement. [Apr/May 2024] Steps 1. Policy Development 2. User Authentication 3. Device Authentication 4. Health Check 5. Authorization 6. Continuous Monitoring Authentication • Password • OTP • Smart Card • Certificate • Biometrics Enforcement • VLAN Assignment • Guest Access • Quarantine • Block Device Diagram Device ↓ Authentication ↓	K4	CO5

S. No	Question	Knowledge Level	Course Outcome
	Policy Check ↓ Authorized / Denied		
2	Device a solution to automate the threat detection system and provide protection against potential threats, in a cloud environment. [Apr/May 2024] Architecture Cloud Users ↓ Traffic Monitoring ↓ Log Collection ↓ AI/ML Threat Detection ↓ IDS/IPS ↓ Response Engine ↓ Administrator Components • Log Collection • SIEM • IDS • IPS • AI Detection • Alert System Features • Real-time Detection • Malware Detection • DDoS Detection • Automated Response Advantages • Fast Response • Reduced Human Effort • High Accuracy	K4	CO5

S. No	Question	Knowledge Level	Course Outcome
3	Analyze the steps involved in SQL injection attack and design a secure login mechanism to prevent it. SQL Injection Steps 1. Identify Input Field 2. Inject SQL 3. Bypass Authentication 4. Access Database 5. Steal Data Secure Login Design Login ↓ Input Validation ↓ Parameterized Query ↓ Password Hashing ↓ Database ↓ Authentication ↓ Dashboard Prevention • Prepared Statements • Password Hashing (bcrypt/Argon2) • MFA • CAPTCHA • Session Timeout • HTTPS	K3	CO5
4	Explain the life cycle of a cyber-attack and how an organization can prepare against it. Cyber Attack Life Cycle Reconnaissance ↓ Scanning ↓ Initial Access	K2	CO5

S. No	Question	Knowledge Level	Course Outcome
	↓ Privilege Escalation ↓ Persistence ↓ Data Exfiltration ↓ Cover Tracks Organization Preparation • Risk Assessment • Security Policies • Employee Training • Incident Response Plan • Regular Backup • Vulnerability Assessment • Patch Management • SIEM • IDS/IPS		
5	Explain cloud, web and wireless security threats with mitigation strategies. Cloud Threats • Data Breach • Insider Attack • API Vulnerability Mitigation • Encryption • IAM • MFA Web Threats • SQL Injection • XSS • CSRF • Session Hijacking Mitigation • Input Validation • WAF • HTTPS • Secure Coding Wireless Threats • Rogue AP • Sniffing	K3	CO5

S. No	Question	Knowledge Level	Course Outcome
	• MITM • Evil Twin Mitigation • WPA3 • VPN • Strong Authentication • Firmware Updates		

COURSE OUTCOMES:

On Completion of the course, the students should be able to:

CO1: Understand the fundamentals of networks security, security architecture, threats and vulnerabilities.

CO2: Apply the different cryptographic operations of symmetric cryptographic algorithms.

CO3: Apply the different cryptographic operations of public key cryptography.

CO4: Apply the various Authentication schemes to simulate different applications.

CO5: Understand various cyber crimes and cyber security.

STAFF INCHARGE

HOD